GONM: A Layered Mathematical and Experimental Framework for Global Optimization on Noisy Multimodal Surfaces

CSD, CPP, MQLM, Robust Terminal Closure, and Dimensional Regimes

Francisco Anderson de Sousa Oliveira

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Abstract

This paper studies GONM as a layered theory of noisy multimodal optimization rather than as a claim of a universal optimizer. The central thesis is that the problem separates into structural basin selection, contractive local refinement, trajectory-level filtering under noise, and, in thin noisy funnels, statistically robust terminal closure. The analysis yields three main conclusions: the benchmark is heterogeneous by regime; the Ackley case reveals a local numerical-statistical barrier that is better handled by repeated-average direct comparison than by further motion heuristics; and dimensional sweeps support a partial structural law for the pure CSD layer before an extreme budget-limited regime emerges. In addition, the layered operator is formalized compositionally and shown to satisfy a qualitative noisy-stability bound under contractive refinement, nonexpansive path filtering, and repeated-average terminal closure. The contribution is therefore a mathematically organized account of stratified difficulty in noisy multimodal optimization.

Keywords: noisy global optimization, multimodal optimization, derivative-free optimization, simulated annealing, structural basin selection, dimensional scaling.

1 Introduction

Global optimization on noisy multimodal surfaces is often treated as a single search problem, whether in classical nonlinear optimization, stochastic search, or derivative-free settings [1, 2, 3, 4]. The analysis developed here suggests that this view is too coarse. More precisely, the difficulty splits into at least four layers:

1. identifying which basins are structurally real and worth preserving,
2. refining candidate basins without amplifying noise,
3. comparing trajectories rather than trusting isolated terminal evaluations,
4. switching to a statistically robust terminal mechanism when the local signal becomes weaker than the local noise floor.

GONM is introduced in order to make this decomposition explicit. In its present form, it is not advanced as a benchmark-universal optimizer. Rather, it is studied as a layered architecture whose behavior clarifies how noisy multimodal optimization organizes itself across different regimes.

The manuscript is organized accordingly. The next sections isolate the mathematical core of the resolution problem, then describe the inherited layered architecture, and only afterwards turn to the experimental evidence.

2 Genealogical Position of GONM

GONM should be read as a synthesis article within the larger sequence of theories developed in this archive, not as an isolated optimizer proposal.

- From `T02_CPP`, it inherits CPP as the contractive stability layer for local refinement.
- From `T03_MQLM`, it inherits MQLM as the trajectory-level logarithmic or multiplicative filter under noisy evaluation.
- From `T04_CSD`, `T05_CSD_V3`, and `T06_CSD_V4`, it inherits CSD as the structural priority principle according to which basin organization should precede local contraction.

What changes in GONM is the mode of use rather than the origin of the components. CPP and MQLM enter as functional layers, whereas CSD enters in a reduced optimization-oriented form adapted to noisy basin discovery under finite budget. The present manuscript is therefore best understood as the architectural synthesis of earlier mathematical lines.

3 Mathematical Setup

I study optimization problems of the form

$$\min_{x \in \Omega} f(x),$$

when only noisy observations

$$y(x) = f(x) + \xi(x)$$

are available, and when the underlying landscape is genuinely multimodal.

The project focused first on low-dimensional diagnostic surfaces and then on dimensional sweeps of `Ackley_d`. The central question was not only which method wins, but also which structural difficulty is actually being encountered in each regime.

4 Assumptions

The mathematical statements below are written under a compact set of standing assumptions.

A1. Noisy oracle. The observable objective has the form $y(x) = f(x) + \xi(x)$, where f is deterministic, $\mathbb{E}[\xi(x)] = 0$, and the evaluation noise has finite variance bounded by σ^2.

A2. Independent repeated calls. Repeated evaluations at a fixed point are independent or weakly dependent in a way that preserves the variance decay of sample averages.

A3. Local smoothness. In local analyses, the objective f is continuously differentiable in a neighborhood of the basin under consideration.

A4. Positive structural hit probability. The structural sampler has positive probability of producing at least one anchor inside a competitive basin under the fixed budget under study.

A5. Admissible regime. The local regime satisfies the admissibility conditions stated later for the layered operator.

A6. Budget density. In dimensional arguments, the effective budget per coordinate may decrease with dimension and can become the dominant source of loss of resolution in the extreme regime.

Assumptions **A1.**–**A3.** support the local resolution statements; Assumptions **A4.**–**A6.** support the layered and dimensional statements.

5 Mathematical Mechanisms

To pass from an algorithmic architecture to a mathematically interpretable one, it is useful to isolate two decisive mechanisms: variance reduction by repeated evaluation and variance amplification in local differential estimates.

Definition 1 (Repeated-evaluation oracle)**.** *Let*

$$y(x) = f(x) + \xi, \qquad \mathbb{E}[\xi] = 0, \qquad \mathrm{Var}(\xi) = \sigma^2,$$

with independent noise across calls. For an integer $m \geq 1$, define the averaged oracle

$$\bar{y}_m(x) = \frac{1}{m} \sum_{k=1}^{m} y_k(x),$$

where $y_k(x)$ are independent evaluations at the same point.

Lemma 1 (Averaging reduces the effective noise floor)**.** *Under the assumptions above,*

$$\mathbb{E}[\bar{y}_m(x)] = f(x), \qquad \mathrm{Var}(\bar{y}_m(x)) = \frac{\sigma^2}{m}.$$

In particular, if two candidate points x_1, x_2 satisfy $f(x_1) - f(x_2) = \Delta > 0$, then the probability that the averaged oracle misranks them decreases monotonically with m.

Proof. Linearity of expectation gives $\mathbb{E}[\bar{y}_m(x)] = f(x)$. Independence gives

$$\mathrm{Var}(\bar{y}_m(x)) = \frac{1}{m^2} \sum_{k=1}^{m} \mathrm{Var}(y_k(x)) = \frac{\sigma^2}{m}.$$

For the difference of two averaged observations,

$$\bar{y}_m(x_1) - \bar{y}_m(x_2) = \Delta + \zeta_m,$$

where ζ_m has zero mean and variance $2\sigma^2/m$. Hence the probability that the sign of the noisy difference disagrees with the sign of Δ decreases as m increases. □

Definition 2 (Noisy forward difference)**.** *For a step size $h > 0$, define the one-sided noisy finite-difference estimator*

$$\widehat{g}_h(x) = \frac{y(x+h) - y(x)}{h}.$$

If each evaluation is itself replaced by an average of m independent calls, write the resulting estimator as $\widehat{g}_{h,m}(x)$.

Lemma 2 (Differential resolution barrier)**.** *Assume again independent centered noise of variance σ^2. Then*

$$\mathbb{E}[\widehat{g}_{h,m}(x)] = \frac{f(x+h) - f(x)}{h}, \qquad \mathrm{Var}(\widehat{g}_{h,m}(x)) = \frac{2\sigma^2}{mh^2}.$$

Consequently, when $h \to 0$ with fixed m, the variance of the differential estimator diverges like h^{-2} even though the deterministic signal converges to $f'(x)$ when the derivative exists.

Proof. The expectation is immediate. Since the two averaged observations are independent and each has variance σ^2/m,

$$\mathrm{Var}(\widehat{g}_{h,m}(x)) = \frac{1}{h^2}\left(\frac{\sigma^2}{m} + \frac{\sigma^2}{m}\right) = \frac{2\sigma^2}{mh^2}.$$

The divergence for $h \to 0$ follows immediately. □

Theorem 1 (Resolution-limited local descent under noisy evaluation)**.** *Assume **A1**–**A3**. In addition, let $x^\star \in \Omega$ be a local minimizer, and let the observation model be*

$$y(x) = f(x) + \xi, \qquad \mathbb{E}[\xi] = 0, \qquad \mathrm{Var}(\xi) = \sigma^2,$$

with independent noise across calls. Let x lie in a neighborhood of $x^\star$ and consider two local inference mechanisms:

1. *a differential mechanism based on $\widehat{g}_{h,m}(x)$ with step size $h > 0$ and m averaged evaluations per point;*
2. *a direct ranking mechanism comparing x with a candidate z through averaged values $\bar{y}_m(x)$ and $\bar{y}_m(z)$.*

If $\|\nabla f(x)\| \to 0$ as $x \to x^\star$, then for fixed m the signal-to-noise ratio of the differential mechanism necessarily collapses as one simultaneously requires local fidelity $h \to 0$. By contrast, for any fixed pair (x, z) with gap $\Delta = f(x) - f(z) > 0$, the ranking error of the direct mechanism can be made arbitrarily small by choosing m sufficiently large.

Proof. Because f is differentiable near x, we have

$$\frac{f(x+h) - f(x)}{h} = \nabla f(x) \cdot u + o(1)$$

along the direction $u = h/\|h\|$. Hence the deterministic signal of the differential mechanism is of order $\|\nabla f(x)\| + o(1)$, while Lemma 2 gives standard deviation

$$\sqrt{\mathrm{Var}(\widehat{g}_{h,m}(x))} = \frac{\sqrt{2}\,\sigma}{\sqrt{m}\,h}.$$

Therefore its signal-to-noise ratio is bounded, up to local constants, by

$$\mathrm{SNR}_{\mathrm{diff}}(x, h, m) \lesssim \frac{\sqrt{m}\, h\, \|\nabla f(x)\|}{\sigma}.$$

As $x \to x^\star$ one has $\|\nabla f(x)\| \to 0$, and if at the same time local fidelity requires $h \to 0$, then $\mathrm{SNR}_{\mathrm{diff}}(x, h, m) \to 0$ for fixed m.

For the direct mechanism, Lemma 1 implies that

$$\bar{y}_m(x) - \bar{y}_m(z) = \Delta + \zeta_m, \qquad \mathrm{Var}(\zeta_m) = \frac{2\sigma^2}{m}.$$

Thus the standard deviation of the comparison noise is $\sqrt{2}\sigma/\sqrt{m}$, while the deterministic separation remains the fixed gap $\Delta > 0$. Hence

$$\mathrm{SNR}_{\mathrm{rank}}(x, z, m) \asymp \frac{\sqrt{m}\,\Delta}{\sigma},$$

which diverges as $m \to \infty$. Equivalently, the ranking error can be made arbitrarily small by increasing m. □

> Theorem 1 gives the manuscript its most general mathematical claim: near a noisy thin minimum, local differential refinement can become resolution-limited even when the model is smooth, whereas short-range direct comparison with repeated averaging remains statistically recoverable.
>
> These two lemmas explain the main diagnostic lesson of the Ackley analysis: repeated averaging improves local ranking exactly in the regime where naive differential information becomes least reliable.

This concludes the mathematical core. The next sections describe its architectural organization inside GONM before turning to the empirical evidence.

6 Architectural Synthesis

Principle 1 (Layered decomposition)**.** *Global optimization on noisy multimodal surfaces should be decomposed into:*

1. *structural identification of competitive basins,*
2. *contractive refinement inside selected basins,*
3. *robust trajectory acceptance under noisy evaluation,*
4. *statistically robust terminal closure near thin noisy minima whenever local differential information becomes unreliable.*

Within GONM, the first three layers are associated respectively with CSD, CPP, and MQLM. The fourth layer arises from the inadequacy of purely differential local closures in thin noisy funnels.

In the present manuscript, ‘CSD‘ should be understood in a reduced optimization-oriented sense. It is inherited from the broader Canonical Structural Decomposition line, but it is not identified here with the full affine, nonlinear single-chart, or multichart formalisms developed in the structural manuscripts. Operationally, GONM uses a simplified structural layer whose task is to discover, preserve, merge, and repopulate competitive basins before local refinement begins. In code, this layer is implemented through population sampling, elite selection, coarse clustering, merging of nearby centers, and structurally weighted repopulation. Thus the role of ‘CSD‘ in GONM is faithful to the structural priority of the CSD line, while remaining a reduced basin-organization mechanism rather than a literal reuse of the full V2/V3/V4 theory.

7 Roles of the Components

CSD. CSD acts as a reduced structural multi-basin organizer. Its role is to preserve several competitive basins, reduce premature collapse, and separate genuine multimodal structure from spurious fragmentation induced by noise through population sampling, elite clustering, center merging, and multibasin repopulation.

CPP. CPP acts as a local stability principle. It favors damped and contractive updates, prioritizing iterative stability over aggressive local motion.

MQLM. MQLM acts as a trajectory-level filter. Rather than trusting only final values, it scores paths through logarithmic or multiplicative aggregation and helps reject unstable improvements that are not structurally convincing.

Robust terminal closure. In thin noisy funnels, gradient information can become numerically weaker than the local noise floor. In that case, a short direct search with repeated local averaging can outperform further gradient-based refinement.

8 Abstract Formulation of GONM

To separate algorithm, hypotheses, and consequence more clearly, it is useful to describe GONM as an abstract layered operator acting on a noisy optimization problem.

Definition 3 (Layered GONM operator)**.** *Fix a noisy objective pair (f, y) on a domain Ω, where y is the observable oracle. A layered GONM operator is a map*

$$\mathcal{G} = \mathcal{C}_{\text{out}} \circ \mathcal{T} \circ \mathcal{M} \circ \mathcal{P} \circ \mathcal{R} \circ \mathcal{C}_{\text{in}} \circ \mathcal{S},$$

where:

1. *$\mathcal{S}$ is a structural sampler producing a finite family of candidate basins or anchors;*
2. *$\mathcal{C}_{\text{in}}$ is a structural classifier selecting the regime associated with each anchor;*
3. *$\mathcal{R}$ is a local contractive refiner acting inside the selected basin;*
4. *$\mathcal{P}$ is a trajectory filter assigning path-level scores under noisy evaluation;*

5. *$\mathcal{M}$ is a regime switch deciding whether differential refinement remains trustworthy;*
6. *$\mathcal{T}$ is a terminal operator, either differential or repeated-average direct search;*
7. *$\mathcal{C}_{\text{out}}$ is the external comparison operator selecting the final candidate.*

Definition 4 (Admissible layered regime)**.** *I call a local regime admissible for the layered operator if the following hold in a neighborhood of the selected anchor:*

a. *the refiner $\mathcal{R}$ is mean-contractive in the sense that expected local displacement decreases the distance to a basin center whenever the local signal dominates noise;*
b. *the path filter $\mathcal{P}$ is monotone with respect to repeated averaged improvement along accepted trajectories;*
c. *the switch $\mathcal{M}$ activates the terminal operator $\mathcal{T}$ once the differential signal-to-noise ratio falls below a fixed threshold;*
d. *the terminal operator $\mathcal{T}$ uses repeated averaging whenever the regime is declared resolution-limited.*

Definition 5 (Componentwise layered error)**.** *Let $x^\star$ be a target local minimizer inside a competitive basin and let X_k denote the random state produced after the k-th active stage of the layered operator. I define the componentwise layered error by*

$$E_k := \mathbb{E}\big[\|X_k - x^\star\|\big],$$

and I split its evolution into three contributions:

a. *a structural transfer error ε_S produced by imperfect basin delivery from $\mathcal{S} \circ \mathcal{C}_{\text{in}}$;*
b. *a contractive local error propagated by $\mathcal{R}$;*
c. *a filtering and terminal error $\varepsilon_{P,T}$ produced by noisy path selection and terminal discrimination.*

Theorem 2 (Qualitative noisy stability of the composed layered operator)**.** *Let $\mathcal{G} = \mathcal{C}_{\text{out}} \circ \mathcal{T} \circ \mathcal{M} \circ \mathcal{P} \circ \mathcal{R} \circ \mathcal{C}_{\text{in}} \circ \mathcal{S}$ be a layered GONM operator. Assume **A1**–**A5**, and suppose in addition that there exist constants $0 < \rho < 1$, $c_P \geq 0$, and $c_T \geq 0$ such that:*

1. *conditioned on entering a competitive basin, the refiner $\mathcal{R}$ satisfies the mean-contractive estimate*
$$\mathbb{E}\big[\|\mathcal{R}(x) - x^\star\| \,\big|\, x\big] \leq \rho\,\|x - x^\star\| + \eta_R,$$
for some local noise term η_R;
2. *the path filter $\mathcal{P}$ is nonexpansive in expectation up to noise, in the sense that*
$$\mathbb{E}\big[\|\mathcal{P}(x) - x^\star\| \,\big|\, x\big] \leq \|x - x^\star\| + \eta_P,$$
with $\eta_P \leq c_P\sigma$;
3. *once $\mathcal{M}$ activates repeated-average terminal closure, the terminal stage satisfies*
$$\mathbb{E}\big[\|\mathcal{T}(x) - x^\star\| \,\big|\, x\big] \leq \|x - x^\star\| + \eta_T(m),$$
where $\eta_T(m) \leq c_T\sigma/\sqrt{m}$.

Then the final expected error of the composed operator satisfies a qualitative bound of the form

$$E_{\text{final}} \leq \varepsilon_S + \rho^q E_{\text{in}} + \frac{1-\rho^q}{1-\rho}\,\eta_R + c_P\sigma + \frac{c_T\sigma}{\sqrt{m}},$$

where q is the number of effective local refinement steps after structural delivery and E_{in} is the expected error at basin entry. In particular, for fixed structural delivery quality, the composed operator is stable in noisy regime whenever $\rho < 1$, and its terminal statistical contribution decays like $m^{-1/2}$ under repeated averaging.

Proof. After structural sampling and classification, the operator either fails to deliver a competitive basin or enters one with expected error E_{in}. The former contribution is recorded as ε_S.

Inside a competitive basin, repeated application of the mean-contractive bound for $\mathcal{R}$ yields by induction

$$E_q^{(R)} \leq \rho^q E_{\text{in}} + \sum_{j=0}^{q-1} \rho^j \eta_R = \rho^q E_{\text{in}} + \frac{1-\rho^q}{1-\rho}\eta_R.$$

The filter $\mathcal{P}$ is nonexpansive up to the additive noise term η_P, so it adds at most $c_P\sigma$ in expectation. Once the regime switch activates repeated-average terminal closure, the terminal contribution is bounded by $\eta_T(m) \leq c_T\sigma/\sqrt{m}$. Summing the structural, contractive, filtering, and terminal contributions gives the claimed bound. □

Corollary 1 (Asymptotic local stabilization under repeated averaging)**.** *Under the assumptions of Theorem 2, if the structural delivery term ε_S remains controlled, if $\rho < 1$, and if the terminal replication count $m \to \infty$, then the final expected error is asymptotically controlled by the structural term together with the bounded accumulated local noise of the contractive stage. In particular, the terminal contribution ceases to dominate the asymptotic error budget.*

Theorem 3 (Layered consequence of the GONM abstraction)**.** *Let (f, y) be a noisy multimodal problem and let $\mathcal{G}$ be a layered GONM operator. Suppose that Assumptions **A4.**, **A5.**, and **A1.**–**A3.** hold, and that the switch $\mathcal{M}$ is consistent in the sense that it activates repeated-average terminal closure whenever the differential signal-to-noise ratio becomes asymptotically smaller than the ranking signal-to-noise ratio of nearby candidates. Then the operator $\mathcal{G}$ has the following consequence: its final decision decomposes into a structural error term and a local statistical resolution term, and in thin noisy funnels the local resolution term is asymptotically lower under repeated-average terminal closure than under purely differential terminal refinement.*

Proof. The structural sampler and classifier determine which basin enters the local stage, so any final error splits into: (i) failure to deliver a competitive basin to the refinement stage, and (ii) failure to resolve the optimum within a basin already judged competitive. This yields the structural term and the local term.

Under admissibility, the refiner $\mathcal{R}$ and the path filter $\mathcal{P}$ do not destroy local information while the differential signal remains usable. Once the differential signal-to-noise ratio falls below the ranking signal-to-noise ratio threshold, consistency of $\mathcal{M}$ forces a switch to the repeated-average terminal operator. Theorem 1 then applies: in the resolution-limited regime, differential inference loses discriminative power whereas repeated-average local ranking remains statistically recoverable. Therefore the asymptotically dominant local error is lower under the switched terminal closure than under a purely differential terminal stage. □

This abstract formulation isolates the central mathematical content of the manuscript: GONM is not merely a list of heuristics, but a layered operator whose benefit comes from separating structural error from local statistical resolution error and assigning a different mechanism to each term.

9 Unified Architecture

Corollary 2 (Hierarchical optimizer)**.** *A natural algorithmic form of GONM is:*

1. *run a structural CSD phase to detect and stabilize competitive basins,*
2. *classify the regime as structural, mixed, or local,*
3. *if local, refine the classification into cleaner local subclasses such as* `local_multiminimo_limpo` *or* `local_agulha_real`*,*
4. *run CPP refinement inside the selected basins,*
5. *use MQLM to rank or accept local trajectories,*
6. *if a thin noisy funnel is detected, switch to a non-gradient robust terminal closure based on repeated local averaging,*
7. *perform a final external comparison between structural and refined candidates.*

This architecture remained more stable under ablation than the fully fused or purely local alternatives examined here.

10 Empirical Support

The remainder of the manuscript is empirical. Its role is to test the layered architecture across low-dimensional diagnostic cases, dimensional sweeps, and a CPU-only physical demonstration.

The manuscript is supported by two complementary computational programs.

Low-dimensional diagnostic benchmark. The 2D suite includes `Rastrigin`, `Ackley`, `Himmelblau`, and a Gaussian mixture landscape under noisy evaluation. These cases isolate periodic multimodality, thin noisy funnels, clean local structure, and pure structural basin selection. They also belong to the family of standard synthetic benchmark landscapes widely used to probe the behavior of global and derivative-free optimization methods [6, 10, 9, 11, 12].

Dimensional sweeps. The second program studies `Ackley_d` from low dimension to very high dimension. This sweep was used to test whether the layered interpretation survives scaling and whether any regularity appears in the structural error of the exploration layer.

11 2D Benchmark Evidence

The low-dimensional benchmark yields a strongly heterogeneous picture.

- On `Rastrigin_2d`, a field-like method remained best on average, which suggests that regular periodic multimodality responds well to broad field exploration.
- On `Ackley_2d`, simulated annealing [5] remained slightly best, but GONM became highly competitive after the introduction of robust terminal closure.
- On `Himmelblau_2d`, clean local contractive methods remained dominant.
- On a Gaussian mixture landscape, structural basin preservation remained dominant, and pure CSD remained best.

This means that the benchmark does not support a single universally dominant optimizer. It supports instead a layered theory of the problem.

Surface	Best method	Best mean	GONM mean
`rastrigin_2d`	phase_field	1.1872×10^{0}	1.7934×10^{0}
`ackley_2d`	sim_anneal	9.2072×10^{-2}	1.0419×10^{-1}
`himmelblau_2d`	fisher_metric	4.3118×10^{-3}	1.1339×10^{-1}
`gaussian_mix_2d`	csd	-6.2522×10^{-1}	-5.8983×10^{-1}

Table 1: Representative results of the 2D noisy benchmark. Lower values are better. The table makes explicit that the strongest method depends on regime, while GONM remains competitive across structurally distinct landscapes.

12 Ackley as the Main Diagnostic Case

Ackley serves as the principal diagnostic case. A large family of local heuristics, acceptance rules, local ontologies, and terminal refiners repeatedly converged to nearly the same value. This indicates not a lack of algorithmic variation, but a local numerical-statistical barrier.

Noise sweeps and finite-difference scale sweeps supported this diagnosis. Near the core of the funnel, the deterministic signal induced by small radial displacements became too weak relative to the noise of the evaluations. The main implication is conceptually important: motion heuristics alone do not break resolution barriers.

A decisive improvement appeared when the terminal closure was changed from gradient-based refinement to a short direct local search with repeated averaging. This does not yield a universal solution to the benchmark, but it shows that the bottleneck lies in local statistical resolution rather than in the choice of a more elaborate geometric motion rule. In that sense, the analysis aligns more closely with derivative-free and noisy direct search methods than with further refinements of a purely differential local closure [3, 4].

13 Dimensional Diagnostic

The first dimensionality study suggested that absolute basin hit rate becomes too brittle as dimension grows. In `Ackley_d`, a more stable observable is the relative residual reduction

$$R = \frac{f_{final} - f^*}{f_{initial} - f^*},$$

computed against a shared initial probe. This quantity remains informative even when fixed hit thresholds become uninformative.

The 10D phase-allocation sweep also showed that GONM benefits from a longer structural handoff in higher dimension. On `Ackley_10d`, allocating roughly 80% of the budget to the structural phase outperformed 52% and 68% under both tested budgets.

These results support a shift in interpretation: as dimension grows, structural allocation becomes a first-order design choice rather than a minor hyperparameter.

14 LJ-38 as a Physical Demonstration

The previous sections focus on synthetic benchmark landscapes. To show that the same layered logic has practical expressive power beyond abstract test functions, we also applied the machinery to a 38-atom Lennard–Jones cluster, a classical nonconvex molecular optimization problem [7, 8]. In this setting the search variable has dimension 114, since each atom contributes three Cartesian coordinates.

The energy is

$$V_{LJ}(x) = \sum_{1 \le i < j \le 38} 4\varepsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^{6} \right],$$

where r_{ij} is the pairwise distance between atoms i and j. This problem is useful here for two reasons. First, it is physically interpretable. Second, it lies in the dimensional range in which structural organization is already important but full asymptotic collapse has not yet taken over.

The LJ-38 computations were carried out in a CPU-only workflow, without specialized accelerators, and therefore provide a concrete demonstration that the proposed architecture can produce substantial geometric improvement on commodity hardware. Table 2 summarizes the main progression observed in the molecular simulations.

Configuration	Evaluations	Runtime	Best final energy
baseline LJ-38	36,336	112.4 s	−15.0152
dense cold start + boundary	203,808	521.6 s	−100.4449
continued refinement, bounds = 4.0	985,104	1885.7 s	−151.1026
parallel escape–freeze (best branch)	$\approx 10^6$ per branch	3080.4 s aggregate	−151.915874

Table 2: Representative LJ-38 results obtained with the Anderson/GONM simulation pipeline on CPU-only execution. The sequence shows a strong reduction in energy as structural seeding, confinement, continuation, and escape–freeze refinement are added.

Two features are especially relevant. First, the move from a naive initialization to a dense cold-start with boundary confinement produces the dominant energy drop, which is consistent with the structural thesis of the manuscript: preserving and compacting the correct basin matters more than aggressive local motion alone. Second, once the search enters the low-energy regime near −151, the returns become marginal and the continuation runs display the same resolution-limited behavior seen in thin-funnel synthetic benchmarks: large additional budgets produce only fine improvements unless a new structural perturbation is introduced.

Figure 1 illustrates the best LJ-38 configuration obtained in this study. Its role is not to claim a new state-of-the-art molecular solver, but to show that the same layered interpretation extends to a genuinely physical, high-dimensional, strongly nonconvex energy landscape.

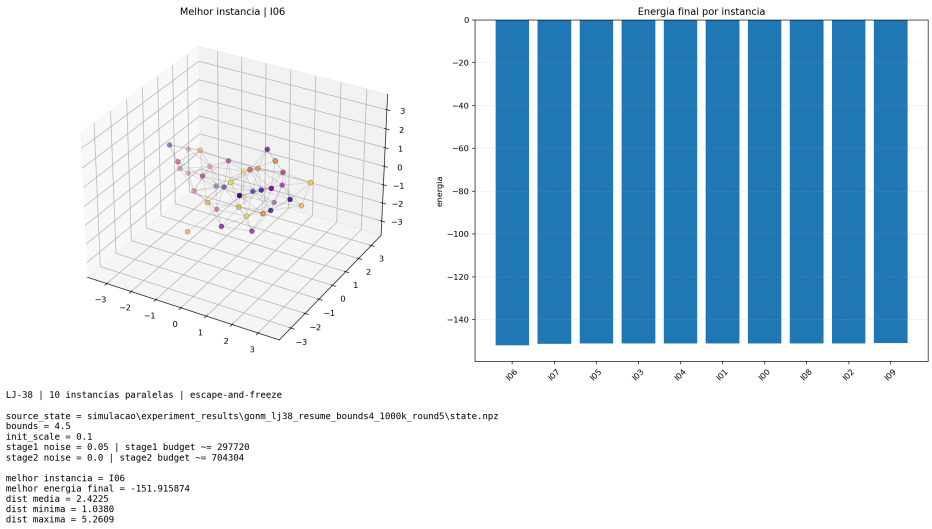

Figure 1: Best LJ-38 configuration obtained by the present CPU-only GONM/Anderson pipeline after continuation and parallel escape–freeze refinement.

15 A Conjectural Structural Law for CSD

The dimensional sweeps provide partial support for a residual wall with effective $\sqrt{D}$ scaling. The effect is not a clean law from 2D onward, but a plateau begins to emerge in medium and high dimensions. The key point is that this regularity appears much more clearly in the pure structural layer than in the full hybrid optimizer.

Conjecture 1 (Partial structural law for CSD)**.** *Consider a family of noisy multimodal problems indexed by dimension D under a fixed evaluation protocol satisfying Assumptions* ***A1.***, ***A4.***, *and* ***A6.***. *In an exploration-dominated dimensional window, the pure structural layer CSD obeys an approximate law of the form*

$$\mathbb{E}[f_{best}(D)] \approx C\sqrt{D},$$

for an effective constant $C > 0$ depending on the problem family and budget density.

I interpret the observed $\sqrt{D}$ wall not as a universal law of GONM, but as a conjectural empirical structural law of CSD in the exploration-dominated regime.

Empirical support. Across the dimensional sweeps, the normalized coefficient $\mathbb{E}[f_{best}]/\sqrt{D}$ exhibits a substantially flatter profile for pure CSD than for the hybrid architecture in the medium/high-dimensional window. The evidence is strongest roughly from $8D$ to $24D$, becomes weaker outside that window, and breaks down in the extreme budget-limited regime. For this reason the manuscript presents the statement as a conjecture supported by evidence rather than as a proved law.

Definition 6 (Structural wall coefficient)**.** *For a fixed problem family in dimension D, define the empirical structural wall coefficient of a method by*

$$W_D := \frac{\mathbb{E}[f_{best}(D)]}{\sqrt{D}},$$

where the expectation is taken over seeds under a fixed evaluation budget.

Corollary 3 (Meaning of the partial $\sqrt{D}$ law)**.** *If W_D remains approximately stable across a dimensional window, then the corresponding method exhibits an error scale of the form*

$$\mathbb{E}[f_{best}(D)] \approx C\sqrt{D}$$

in that window for some effective constant $C > 0$.

Proof. The statement is a direct rearrangement of the definition. Approximate stability of W_D means precisely that there exists an effective constant C such that $W_D \approx C$, and therefore $\mathbb{E}[f_{best}(D)] \approx C\sqrt{D}$ in the observed regime. □

Theorem 4 (Budget, dimension, and loss of local resolution)**.** *Assume **A1**–**A6**. Let $B(D)$ denote the total evaluation budget available in dimension D, let $m(D)$ denote the effective number of repeated evaluations used in the terminal local comparison stage, and suppose that the local differential stage uses a fidelity scale $h(D) \to 0$ as $D \to \infty$. If*

$$\frac{B(D)}{D}$$

remains bounded and the effective replication count $m(D)$ does not grow fast enough to compensate for the decay of the local signal, then there exists an extreme-dimensional regime in which both of the following occur:

1. *the differential local signal-to-noise ratio collapses uniformly over the terminal neighborhood;*
2. *increasing the structural share of the budget yields diminishing returns because the experiment becomes dominated by budget density rather than by basin ranking quality.*

Proof. By Theorem 1, the local differential signal-to-noise ratio scales like

$$\mathrm{SNR}_{\mathrm{diff}}(D) \lesssim \frac{\sqrt{m(D)}\, h(D)\, s(D)}{\sigma},$$

where $s(D)$ denotes the characteristic local signal magnitude in the terminal neighborhood. If $h(D) \to 0$ and $m(D)$ does not grow sufficiently fast relative to the decay of $s(D)$, then this quantity collapses. That proves the first claim.

For the second claim, Assumption **A6.** states that budget density can become the dominant limitation. Once $B(D)/D$ remains bounded, each coordinate receives only bounded effective resolution. In that setting, further reallocating a fixed total budget toward the structural phase improves basin ranking only weakly compared with the loss incurred by insufficient local resolution. Therefore the marginal gain of increasing the structural share must eventually diminish, yielding an extreme budget-limited regime. □

> This theorem is global and qualitative: it does not claim a universal critical dimension, but it does formalize the mechanism behind the transition from a structural scaling regime to an extreme budget-limited regime.

16 Two Asymptotic Regimes

The dimensional evidence is better described by two asymptotic regimes rather than one uniform high-dimensional law.

Regime I: medium/high-dimensional structural regime. Roughly from $8D$ to $24D$, the structural CSD layer obeys the partial $\sqrt{D}$ law described above. In this regime, GONM operates systematically below the structural wall, which suggests that the hybrid architecture acts as a correction over the exploration barrier rather than merely reproducing it.

Regime II: extreme budget-limited regime. At very high dimension, such as $240D$ and $1000D$, increasing the structural share no longer changes the outcome appreciably. The previous $\sqrt{D}$ plateau no longer extends in a simple way. The analysis becomes dominated by budget per coordinate, and the problem enters a budget-limited regime.

Thus, the partial structural law for CSD should be understood as valid only inside the first asymptotic regime, not across the whole dimensional spectrum.

Dimension	sim_anneal	CSD	best GONM	Regime
1	5.1006×10^{-3}	2.3997×10^{-2}	2.2277×10^{-2}	low/local
10	4.1671×10^{0}	3.6904×10^{0}	2.9500×10^{0}	medium/high
100	1.0018×10^{1}	6.5552×10^{0}	5.6458×10^{0}	medium/high
1000	1.0374×10^{1}	7.2442×10^{0}	5.9399×10^{0}	extreme

Table 3: Representative points of the dimensional sweep from 1D to 1000D on `Ackley_d`. The GONM column reports the best tested structural policy at each representative dimension.

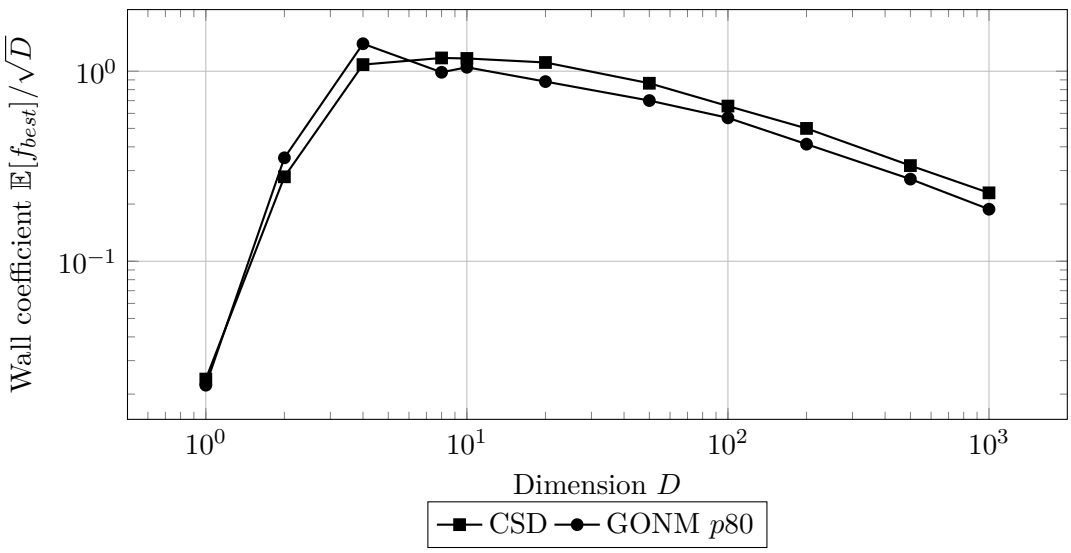

Figure 2: Wall coefficient across representative dimensions. The medium/high-dimensional plateau is much clearer for CSD, while the hybrid architecture remains systematically below the structural wall and eventually enters the extreme budget-limited regime.

17 Proof Status

For clarity, the mathematical status of the claims in this manuscript is divided into three levels.

Proved statements. The local noise-averaging mechanism is proved in Lemma 1, the differential resolution barrier is proved in Lemma 2, the local resolution-limited descent statement is proved in Theorem 1, the composed-operator stability estimate is proved in Theorem 2, the

asymptotic local stabilization statement is recorded in Corollary 1, and the layered qualitative consequence is formalized in Theorem 3. The extreme-regime mechanism linking budget density, dimension, and loss of local resolution is formalized in qualitative form in Theorem 4.

Derived consequences. The hierarchical algorithmic form of the method is presented as Corollary 2, and the interpretation of an approximately flat wall coefficient as an effective $\sqrt{D}$ scaling law is recorded in Corollary 3.

Conjectural and empirical statements. The dimensional scaling statement for the pure structural layer is not presented as a theorem. It is stated as Conjecture 1 and supported by the dimensional sweeps, tables, and figures. The division into medium/high-dimensional and extreme budget-limited regimes is likewise supported empirically, although Theorem 4 gives a mathematical mechanism compatible with that transition.

This separation is intentional. The manuscript distinguishes rigorously proved local mechanisms from higher-level structural claims that are conjectural but empirically supported.

18 Contributions

The manuscript makes five distinct contributions.

1. It proposes a layered interpretation of noisy multimodal optimization in which structural selection, local contraction, trajectory filtering, and terminal statistical closure play different mathematical roles.
2. It formalizes the composed layered operator and derives a qualitative noisy-stability bound separating structural, contractive, filtering, and terminal contributions to the final expected error.
3. It identifies Ackley as a diagnostic example of a local numerical-statistical barrier and formalizes why repeated averaging can help precisely where local differential estimation becomes unreliable.
4. It documents a dimensional transition in which the pure structural layer exhibits a partial $\sqrt{D}$ law in a medium/high-dimensional regime.
5. It isolates an extreme budget-limited regime in which the previous scaling picture no longer extends in a simple way.

19 Findings

The analysis supports five high-level findings.

1. Low dimension is not yet asymptotic: the range from 1D to 4D remains dominated by local behavior and irregular transitions.
2. The range from roughly 5D to 100D behaves as a coherent medium/high-dimensional regime in which structural organization becomes increasingly important.

3. The first law-like regularity appears at the level of the structural layer: CSD exhibits a partial $\sqrt{D}$ law in the medium/high-dimensional regime.
4. GONM remains systematically below the structural wall of CSD, which suggests that the hybrid architecture acts as a correction over the structural barrier rather than merely reproducing it.
5. Extreme dimension introduces a different asymptotic regime: at 240D and 1000D, the problem becomes budget-limited and increasing the structural share no longer changes the outcome appreciably.

These findings imply that the main contribution of the present line is not the discovery of a universally dominant optimizer, but the identification of a stratified difficulty structure for noisy multimodal optimization.

20 Limitations

The present manuscript remains limited in scope.

- It does not provide a general convergence theorem for the full architecture.
- The strongest claims are empirical and architectural rather than axiomatic.
- The low-dimensional suite is informative but not exhaustive.
- The dimensional law is partial and belongs more clearly to CSD than to GONM as a whole.
- The extreme-dimensional regime still lacks a satisfactory theory beyond the budget-limited diagnosis.

For these reasons, the present contribution should be read as a layered theory with rigorous local components and conjectural higher-level structure, rather than as a closed mathematical theory of global noisy optimization.

21 Conclusion

GONM should not be interpreted as a claim that one optimizer can dominate every noisy multimodal benchmark. Its contribution is different. The analysis supports the view that noisy multimodal optimization is stratified: structural basin selection, stable local contraction, trajectory-level filtering, and, in the thinnest noisy funnels, statistically robust terminal closure play distinct roles and must not be collapsed into a single mechanism.

The conclusion is therefore methodological as much as algorithmic. Heuristics of motion alone do not resolve local resolution barriers. Once the useful local signal falls below the local noise floor, progress depends less on altering the step policy than on altering the mode of inference itself. Read in the context of the archive, GONM is the synthesis of the previous line: `T02` supplies the contractive local layer, `T03` the trajectory filter, and `T04`–`T06` the structural priority principle. The CPU-only LJ-38 results show that this synthesis is not confined to synthetic benchmarks, since on ordinary hardware it produced a substantial and reproducible reduction of molecular energy in a genuinely high-dimensional physical landscape.

Declarations

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Conflicts of interest The author declares that there is no conflict of interest.

Data availability The experimental results discussed in this manuscript are stored in the project repository, including the benchmark outputs under `out/` and `GONM/results/`.

Code availability The implementation used in this study is available in the project repository, including the main benchmark scripts and the supporting simulation code, under the GNU AGPLv3 license.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

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